AUCKLAND INTERNATIONAL CAMPUS | FutureSkills | OTAGO POLYTECHNIC Te Kura Matatini ki Otago

IA736001 Internet of Things and Cloud Computing

Project: Design and Develop a Cloud-based IoT Solution

Study Block:	Block 3, 2025: 28 July 2025 to 26 September 2025
Course Leader:	Dr Senaka Amarakeerthi
Due date and time:	17th September 2025
Submission:	Moodle Submission
Submission format:	Project portfolio, Video recordings, Presentation/demo materials (e.g. slides), Any other complementary documents as evidence for your portfolio (If you are willing to get the assistance of AI, please submit your original document as well) (Read Responsible Use of AI guidelines)
Weighting/Contribution:	60%
Marks out of	100
Learning outcomes covered:	LO 1, LO2 and LO3
Version	1.0

Assessment:

For this part of the assessment, you will implement the project you explained in the Case Study Analysis. This project should be supported by the **evidence you will provide,** including documented code, video, and photographic evidence. The project should demonstrate a coverage and depth of research on the following knowledge domains:

- Networking Protocols
- Hardware Platforms
- Software and Applications
- Cloud platform
- Security

Project Deliverables:

You will be required to **demo and defend** your working prototype. The total presentation time will be 10 minutes plus 5 minutes for questions and answers.

You will also create a **portfolio document** including sufficient description of each part as mentioned in the instructions above, pictures/screenshots, figures, code snippets, etc. Demo videos could be submitted to OneDrive, and the link provided in the portfolio. The document should be created with professional quality, structure, writing, and formatting.

For this part of the project, submit **the following documents** to Moodle using the submission link provided for this assessment in the course Moodle page:

- Project portfolio
- Presentation/demo materials (e.g., slides)
- Any other complementary documents as evidence for your portfolio (as described in Instructions)

  You must also submit an appropriate cover sheet for the assessment.

Assessment Criteria and Marking Scheme

Criteria	80%–100% (Worthy of top marks)	65%–79% (Better than average)	50%–64% (Enough for a pass)	30%–49% (Struggling)	0%–29% (Insufficient)	Marks
Project Implementation (Functional Prototype)	Project is fully functional, demonstrates strong technical skills and creativity; aligns completely with case study.	Project is functional with minor issues; aligns well with case study.	Project is partially functional; some objectives achieved.	Project shows limited functionality; significant gaps in requirements.	Project is non-functional or missing.	30
Coverage of Knowledge Domains (Networking, Hardware, Software, Cloud, Security)	All five domains are comprehensively covered with clear evidence and strong technical justification.	Four domains covered in sufficient detail.	Three domains reasonably covered.	One or two domains covered with limited detail.	Domains missing or coverage is minimal.	25
Portfolio Quality (Structure, Evidence, Professionalism)	Portfolio is well-structured, professionally formatted; includes clear descriptions, screenshots, figures, code snippets, and working video evidence.	Portfolio is structured and includes most required elements; evidence and formatting generally clear.	Portfolio has basic structure; limited evidence and formatting issues.	Portfolio poorly structured; minimal evidence included.	Portfolio missing or does not meet requirements.	20
Presentation & Defense (Demo + Q&A)	Presentation is highly professional, clear, and engaging; excellent demo and confident, well-informed responses in Q&A.	Presentation is clear and organized; responses to questions are satisfactory.	Presentation lacks clarity in parts; responses to Q&A are basic.	Presentation is unclear or incomplete; weak Q&A responses.	No presentation or extremely poor effort.	15

Documentation & Supporting Materials (Slides, Code, Links)	Complete, well-prepared slides and supporting materials; code is clearly documented; video links accessible and functional.	Slides and materials mostly complete; code partially documented; video links provided but minor issues.	Basic slides and materials; code documentation limited; video links unclear or incomplete.	Incomplete slides/materials; missing documentation; video evidence insufficient.	No supporting materials provided.	10